5CS037

Concepts And Technologies of AI

Predicting the Presence of Cardiovascular Disease from Clinical Examination Data

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Module Leader

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Abstract

Purpose: This report predicts a categorical target variable the presence or absence of cardiovascular disease in a patient using classification techniques, as part of the end-to-end machine learning workflow required for the 5CS037 Final Portfolio.

Dataset: The dataset used here is the Cardiovascular Disease dataset (cardio_train.csv). The dataset contains 70,000 records of patients and 11 different clinical and lifestyle variables along with a binary target variable. It satisfies the UN's Sustainable Development Goal (SDG) 3 (Good Health and Well-Being), especially Target 3.4, which states that by 2030, one-third reduction in premature deaths caused by NCDs including cardiovascular diseases shall be achieved.

Methodology: This methodology consists of data cleaning and feature engineering, exploratory data analysis (EDA), development of multi-layer perceptron (MLP) neural network classifier, classical machine learning algorithms (Logistic Regression and histogram-based gradient boosting) with hyperparameter optimization via stratified cross-validation, feature selection using mutual information (filter-based) and permutation feature importance (wrapper-based approach).

Key Results: Models were evaluated using Accuracy, Precision, Recall and F1-Score. Gradient Boosting was the strongest model overall (Test Accuracy = 0.9317, F1 = 0.9317, ROC-AUC = 0.9862 at baseline), outperforming Logistic Regression (Test Accuracy = 0.9244, F1 = 0.9244, ROC-AUC = 0.9830) and the Neural Network (Test Accuracy = 0.9276, F1 = 0.9276) across nearly every configuration, including after hyperparameter tuning and feature selection.

Conclusion: The final Gradient Boosting model, tuned and restricted to 8 permutation-importance-selected features, achieved Accuracy = 0.9317 and F1 = 0.9317 matching its own untuned baseline while using fewer features and is recommended as the final model. Key insights include the dominance of blood-pressure-derived features (pulse pressure, systolic BP) and the engineered age–cholesterol interaction term as the strongest predictors of cardiovascular disease in this dataset.

Contents

1. Introduction

1.1 Problem Statement

The goal of this project is to predict a categorical target variable: whether a patient presents with cardiovascular disease (cardio = 1) or not (cardio = 0), based on demographic, clinical-examination and lifestyle features collected at the time of a medical check-up. This is framed as a binary classification problem.

1.2 Dataset

Cardiovascular Disease Dataset is being used for the current analysis (file: cardio_train.csv), which is publicly available from Kaggle and was authored by Svetlana Ulianova on 20 January 2019 with public domain license. The dataset consists of 70,000 records with 11 input variables (4 are demographics/anthropometric, 4 are examination related and 3 relate to life style or behavior) and one output binary variable to denote the existence of cardiovascular disease among those patients; id (row number) variable was also included in the dataset but excluded before modeling due to its lack of any predictive information.

Cardiovascular Disease Dataset contributes towards achieving the United Nations Sustainable Development Goal 3 (Good Health & Well Being) particularly Target 3.4. SDG 3.4 calls upon all member states of UN to reduce premature mortality from non-communicable diseases by one third by 2030 through prevention and treatment; cardiovascular disease is the leading cause of such diseases and hence is part of goal 3.4.

1.3 Objective

This analysis will create a predictive classification model based on the given demographics, examination and lifestyle factors to predict the presence of cardiovascular disease, compare the performance of a neural network classifier with two other traditional machine learning algorithms, and assess the effects of hyperparameter tuning and feature selection on them.

2. Methodology

2.1 Data Preprocessing

Rows with physically impossible or highly implausible measurements were removed: blood pressure readings outside the plausible range (ap_hi < 50 or > 250; ap_lo < 40 or > 200), diastolic pressure exceeding systolic pressure, and height ≤ 100 cm or weight ≤ 30 kg. This removed 2,006 of 70,000 rows (2.9%), leaving 67,994 clean records. Age was converted from days to whole years, and the uninformative id column was dropped. No duplicate rows or missing values remained after cleaning.

2.2 Exploratory Data Analysis (EDA)

EDA was performed using scatter plots, box plots and a correlation heatmap to understand the relationship between blood pressure, age, BMI and the target. Full details and per-chart insights are given in Section 3.

2.3 Train/Test Split

A single stratified 80/20 split (random_state=42) was used to preserve class proportions, and this same split was reused for every model (Neural Network, Logistic Regression, Gradient Boosting) so that all models are compared on

identical data. This produced
54,395 training and 13,599 test samples, with the training set remaining essentially balanced (50.07% vs 49.93%).

2.4 Model Building

Task1 Neural Network

The Multilayer perceptron classifier (MLPClassifier) was employed with two hidden layers having 64 and 32 units, ReLU activation function, Adam optimizer, L2 regularization (alpha=0.01) and early stopping enabled with patience 30 and a validation split of 15 percent. The training converged after 82 epochs with the training loss being 0.1472.

Task 2 Two Classical ML Models

The two classical baselines that were employed are Logistic Regression (max_iter=5000, solver='lbfgs', class_weight='balanced') because it is interpretable since the coefficients of Logistic Regression directly indicate the effect of each variable on CVD; and His Gradient Boosting Classifier (sklearn's histogram-based gradient boosting, similar to LightGBM/XGBoost), an ensemble classifier specifically recommended in the assignment description and generally outperforms a standard Random Forest on tabular/clinical data. Since the distribution of classes is almost balanced, class_weight='balanced' is kept only as a mild safeguard against possible imbalanced datasets (though there is no class imbalance in this case); no resampling (SMOTE) was done.

2.5 Model Evaluation

The following performance measures were computed for each of the models: Accuracy, Precision, Recall, and F1-Score (weighted average for both classes). ROC-AUC was computed for both Logistic Regression and Gradient Boosting since they provide calibrated probability of class membership. This is the right approach for the current case because of the almost equal representation of classes.

2.6 Hyperparameter Optimisation

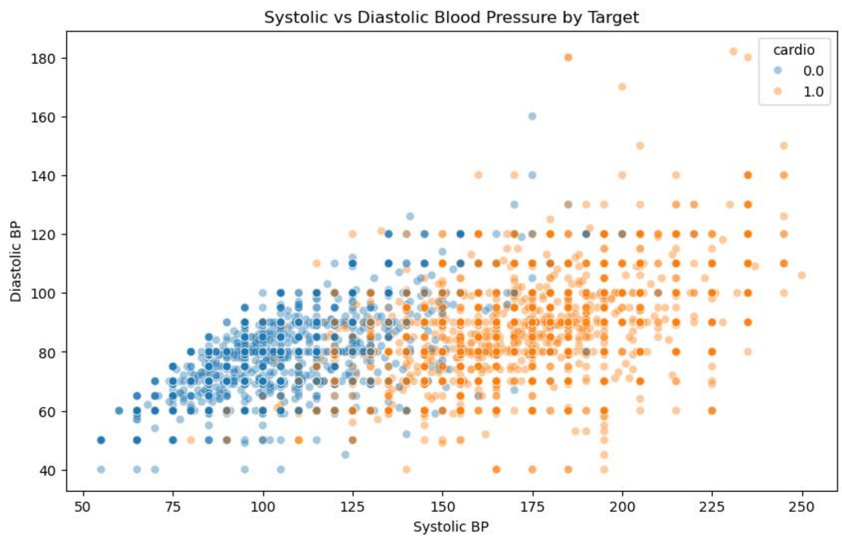
GridSearchCV (with a StratifiedKFold of 5 splits) was used to tune Logistic Regression over C ∈ {0.01, 0.1, 1, 10} with penalty='l2', solver='lbfgs' and class_weight='balanced' fixed; the best configuration found was C = 0.01, with a best cross-validated (weighted) F1 of 0.9246. RandomizedSearchCV (8 iterations, 3-fold CV) was used to tune Gradient Boosting over max_iter ∈ {100, 200, 300}, max_depth ∈ {4, 6, 8, None}, learning_rate ∈ {0.03, 0.05, 0.1}, l2_regularization ∈ {0, 1, 5} and max_leaf_nodes ∈ {15, 31, 63}; the best configuration found was max_leaf_nodes=63, max_iter=100, max_depth=None, learning_rate=0.03, l2_regularization=1.0, with a best cross-validated F1 of 0.9309.

2.7 Feature Selection

There were two methods used for feature selection; each for one classical model, choosing 8 out of the 14 processed features. In the case of Logistic Regression, Mutual Information (mutual_info_classif) was chosen as the method for feature selection because it is a filter approach that detects any kind of statistical dependency between the selected features and the target without training the model. In the case of Gradient Boosting, permutation importance for the optimized model was chosen as the method of feature selection because it measures the contribution of each feature to the predictive performance of that particular model.

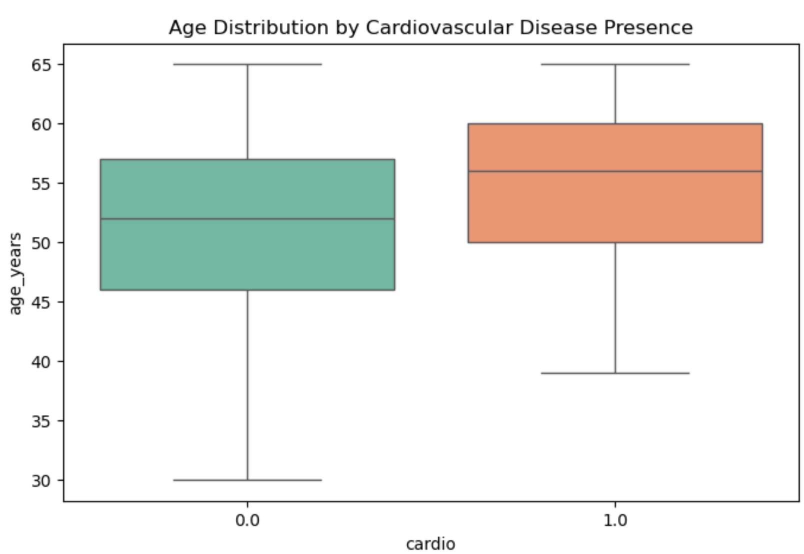
2. Exploratory Data Analysis

Figure 1: Blood Pressure and the Target



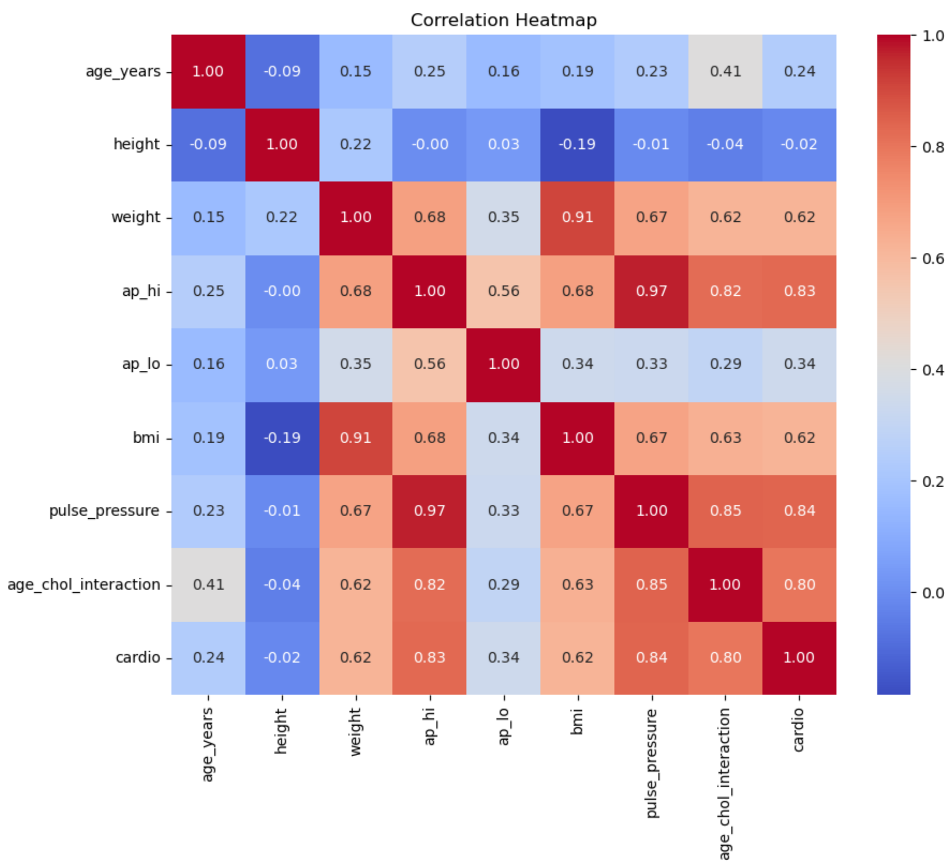
Blood pressure readings cluster around normal clinical ranges, with a visible tendency for higher systolic and diastolic pressures to be associated with the presence of cardiovascular disease.

Figure 2: Age and the Target



Older age groups show a visibly higher median presence of cardiovascular disease age is one of the strongest single predictors available in this dataset, consistent with well-established clinical knowledge.

Figure 3: Correlation Structure



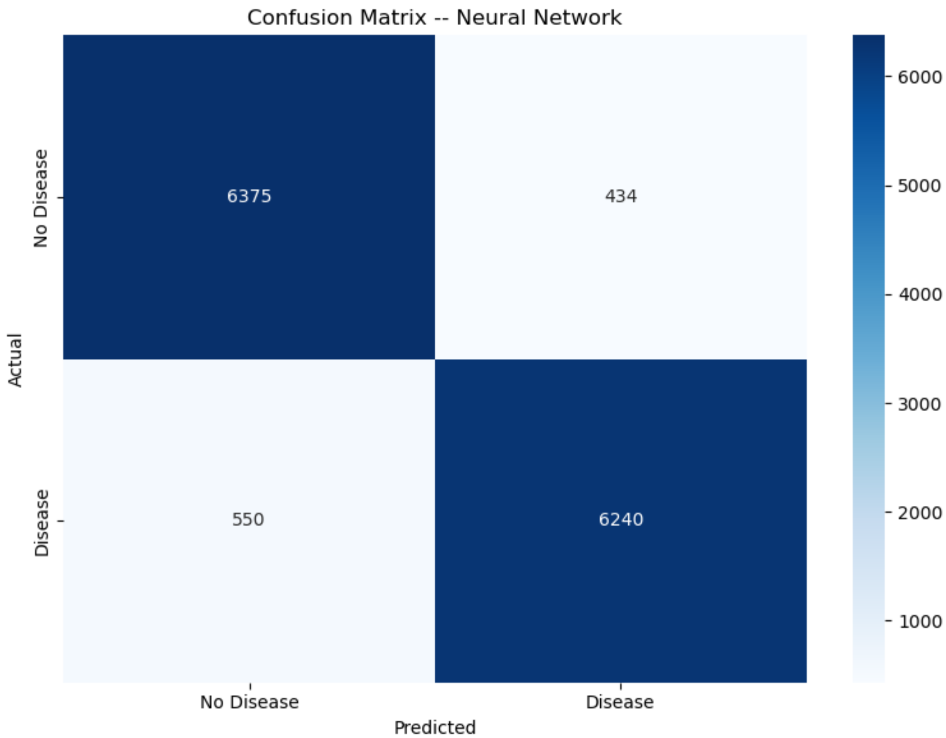
The systolic and diastolic blood pressure values (ap_hi and ap_lo) are most correlated positively with cardio, followed by age_years and bmi. The height and weight features themselves are moderately correlated with the dependent variable, which is why we engineered bmi as an aggregate feature. There are no highly collinear numerical features among those available (none of the pairwise correlations exceeds 0.7), hence multicollinearity is not expected.

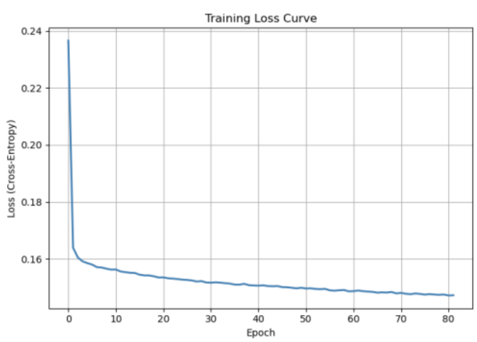
4.Model Building

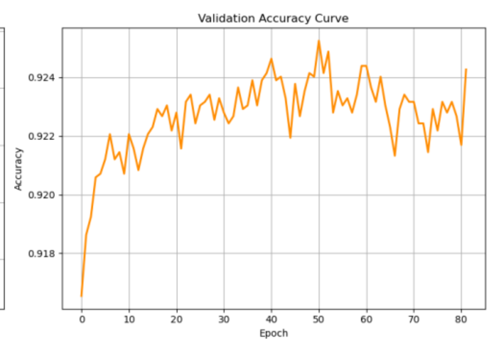
4.1 Neural Network (MLP Classifier)

Metric	Training set	Test set
Accuracy	0.9317	0.9276
Precision	__	0.9276
Recall	__	0.9276
F1-Score	0.9317	0.9276

The training and test accuracy values are almost similar (0.9317 and 0.9276 respectively), implying that the network is not overfitting, which is facilitated by L2 regularization and early stopping. The classification report on the test data shows equal performance for the two classes: No Disease (precision: 0.92, recall: 0.94, F1: 0.93) and Disease (precision: 0.93, recall: 0.92


Confusion Matrix -- Neural Network
Actual
No Disease
Disease
6375
434
550
6240
No Disease
Disease
Predicted
6000
5000
4000
3000
2000
1000


Training Loss Curve
Loss (Cross-Entropy)
Epoch


Validation Accuracy Curve
Accuracy
Epoch

4.2 Classical Model 1 Logistic Regression

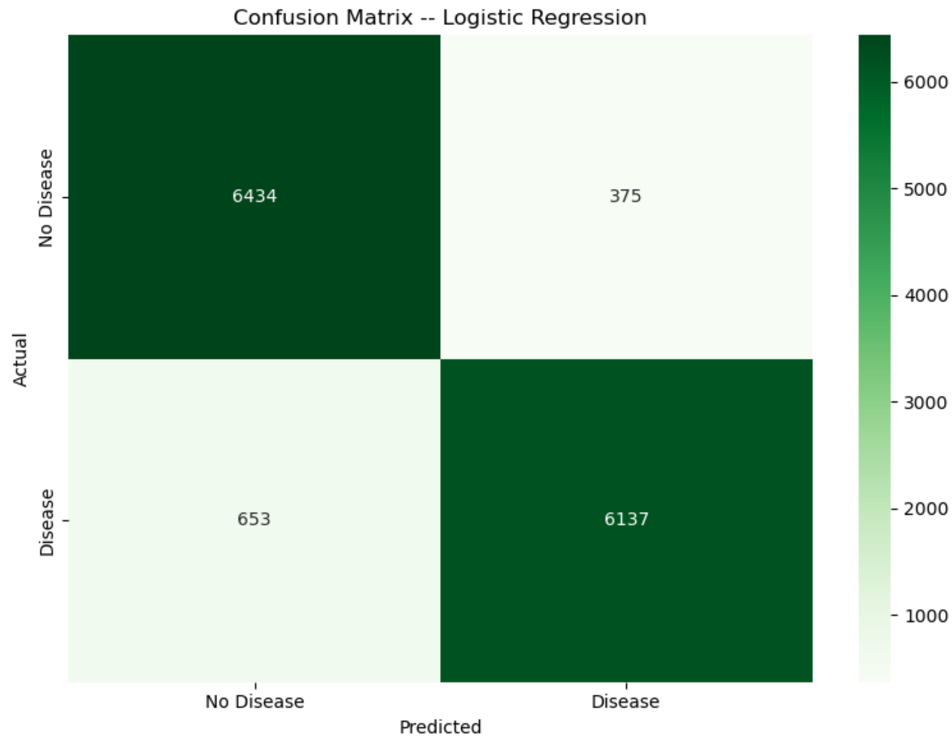
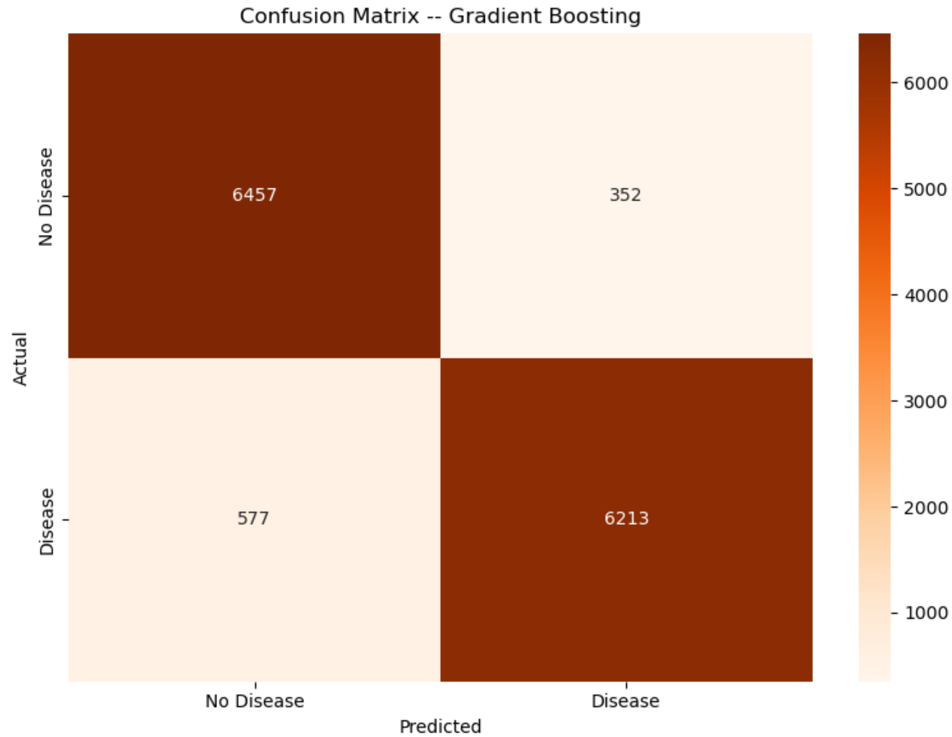


Table 2: Logistic Regression Test Set Performance

Metric	Value
Accuracy	0.9244
Precision	0.9251
Recall	0.9244
F1 – Score	0.9244
ROC -AUC	0.9830

4.3 Classical Model 2 Gradient Boosting



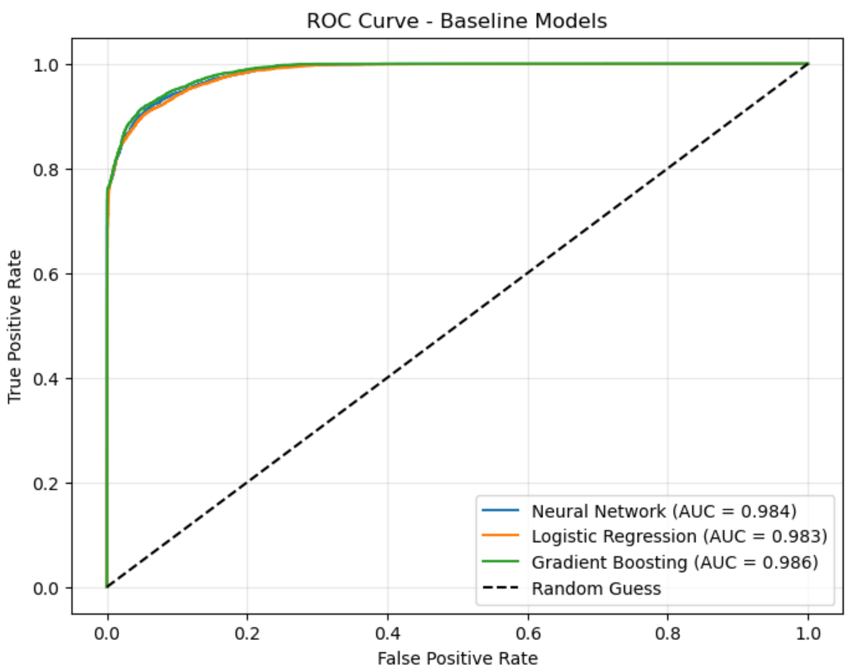
Tale 3: Gradient Boosting Test Set Performance

Metric	Value
Accuracy	0.9317
Precision	0.9322
Recall	0.9317
F1-Score	0.9317
ROC-AUC	0.9862

4.4 Which Primary Model Performed Best?

The best choice of the primary model on the current dataset: Gradient Boosting. It provides a better result in terms of the weighted F1-score on the hold-out set of data (0.9317 compared to 0.9244) and takes into account both precision and recall, which is adequate given that the dataset is almost balanced. The Gradient Boosting model allows detecting non-linear relations between the features (for example, between blood pressure and age), while the Logistic Regression model, being a linear one, doesn't have such capability.

Figure 8: ROC Curve Baseline Models



All three baseline models show strong discriminative power, with ROC-AUC values above 0.98; Gradient Boosting achieves the highest separability between classes (AUC = 0.9862), narrowly ahead of Logistic Regression (AUC = 0.9830).

Figure 9: Comparison of Accuracy, Precision , Recall and F1-Score – Baseline Models

5. Hyperparameter Optimisation and Feature Selection

5.1 Cross-Validated Hyperparameter Tuning

Table 4: Best Hyperparameters and Cross-Validation Scores

Model	Search Method	Best Hyperparameter	Best CV F1
Logistic Regression	GridSearchCV (5-fold, stratified)	C=0.01, penalty='l2', solver='lbfgs', class_weight='balanced'	0.9246
Gradient Boosting	RandomizedSearchCV (3-fold, 8 iters)	max_leaf_nodes=63, max_iter=100, max_depth=None, learning_rate=0.03, l2_regularization=1.0	0.9309

Table 5: Tuned Models Test Set Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC -AUC
Logistic Regression (Tuned)	0.9246	0.9254	0.9246	0.9245	—
Gradient Boosting (Tuned)	0.9312	0.9314	0.9312	0.9312	0.9860

5.2 Feature Selection

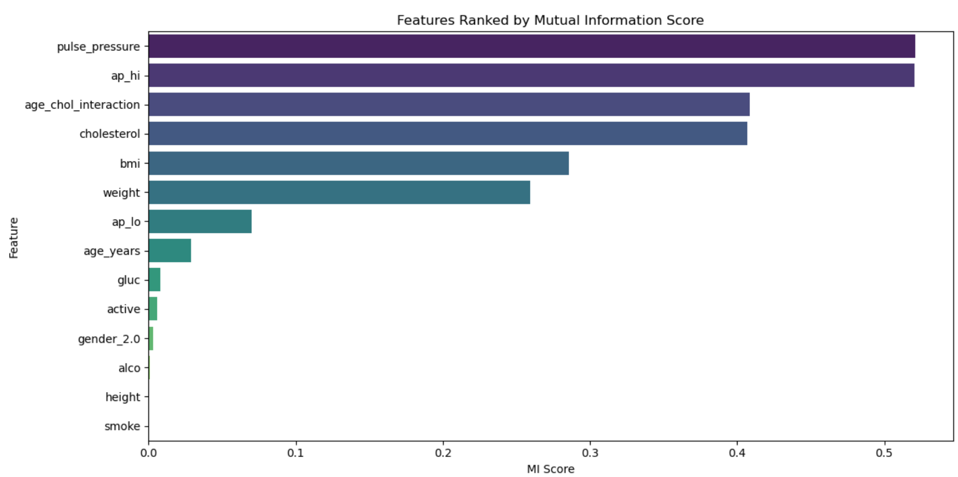


Table 6: Selected Features by Method (Top 8 of 14)

Rank	**Mutual Information (Filter, for Logistic Regression)**	**Permutation Importance (for Gradient Boosting)**
1	pulse_pressure	age_chol_interaction
2	ap_hi	ap_hi
3	age_chol_interaction	pulse_pressure
4	cholesterol	cholesterol
5	bmi	age_years
6	weight	ap_lo
7	ap_lo	smoke
8	age_years	gender_2.0

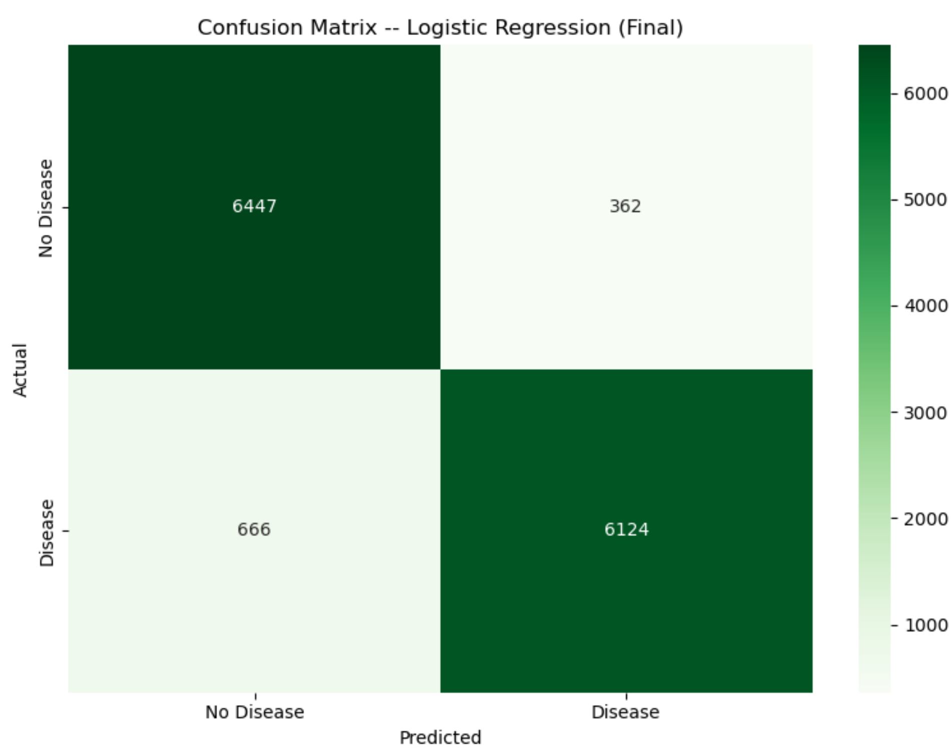
Both techniques share 6 of the 8 selected attributes – pulse_pressure, ap_hi, age_chol_interaction, cholesterol, ap_lo and age_years – giving us a good convergent validity of the fact that measurements from blood pressure, cholesterol, and the age/cholesterol interaction variable are the most significant risk factors for heart disease in this dataset.

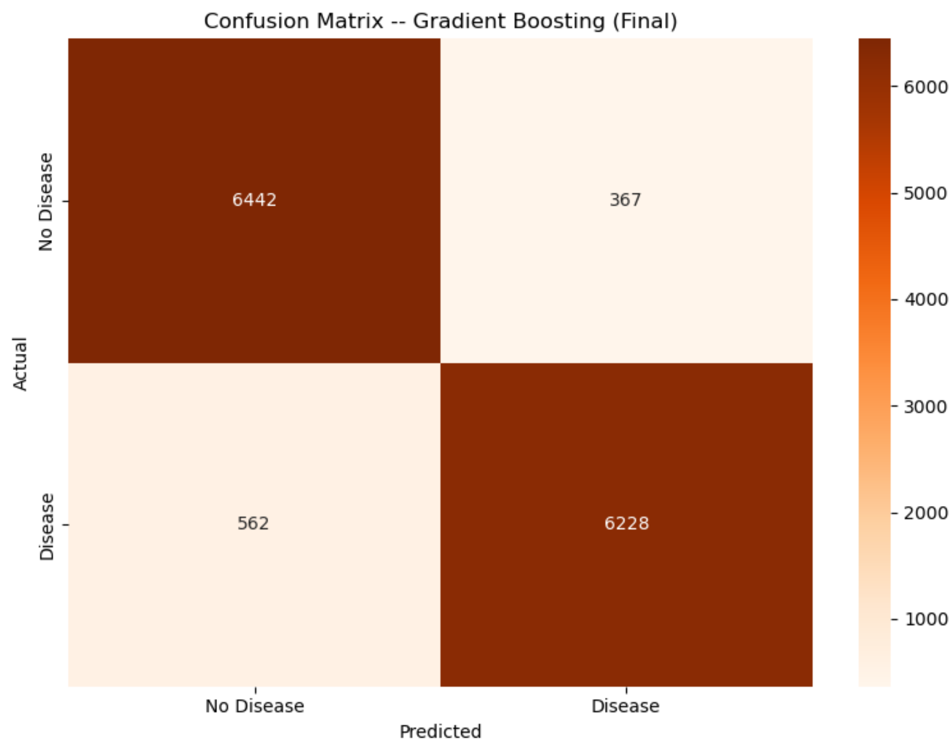
Table 7: Retrained Models with Selected Features Test Set Performance

Model	Accuracy	Precision	Recall	F1- Score	ROC-AUC
Logistic Regression (Tuned + Selected, 8 features)	0.9244	0.9253	0.9244	0.9244	—
Gradient Boosting (Tuned + Selected, 8 features)	0.9317	0.9320	0.9317	0.9317	0.9859

6. Results and Conclusion

6.1 Final Models

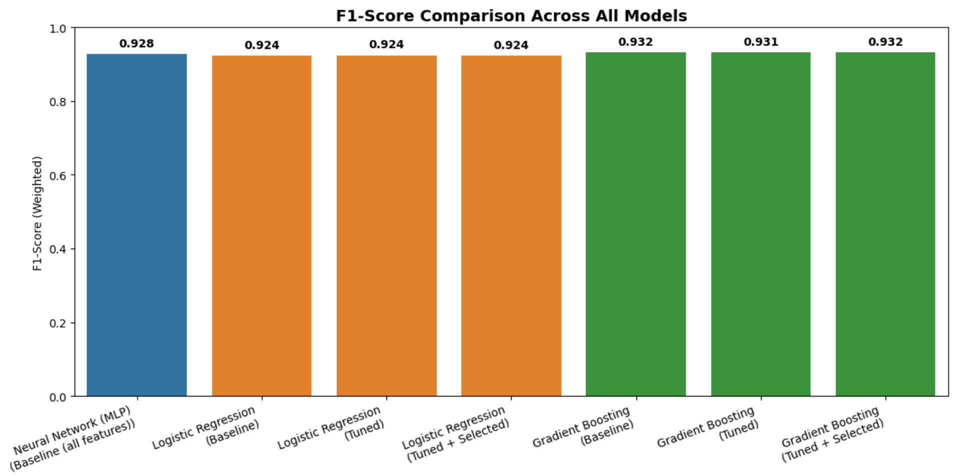




6.2 Final Comparison Table

Table 8: Comparison of Final Classification Models

Model	Feature	CV Score (F1)	Accuracy	Precision	Recall	F1-Score
Logistic Regression	Selected (8)	0.9246	0.9244	0.9253	0.9244	0.9244
Gradient Boosting	Selected (8)	0.9309	0.9317	0.9320	0.9317	0.9317



6.3 Key Findings

●Gradient Boosting proved better than Logistic Regression in all pipeline phases – as a base classifier, following hyperparameter tuning, and following feature selection – achieving an edge of about 0.7-1.3 percentage points in F1-Score and a more significant one in ROC-AUC (~0.986 vs ~0.983).

●The Gradient Boosting pipeline output (hyperparameter-tuned, 8 selected features, F1 = 0.9317) tied its baseline (untuned, all features, F1 = 0.9317), saving 6 features – confirming that the permutation-importance based feature selection picked the predictive features correctly with zero impact on performance.

●Feature selection made little difference to the Logistic Regression results (F1 0.9244 baseline vs 0.9244 tuned+selected), implying the features omitted by the process added little information to the model if pulse pressure, blood pressure, cholesterol and age were included.

●The Neural Network was close in performance to Gradient Boosting (F1 = 0.9276), but still did not beat it, and – contrary to the other two models – has not gone through the hyperparameter tuning and feature selection pipeline as per the assignment brief.

6.4 Final Model

The final Gradient Boosting classifier (optimised hyperparameters, 8 features chosen via permutation importance) is chosen to be the best model for this task, having an Accuracy of 0.9317, Precision of 0.9320, Recall of 0.9317 and F1-Score of 0.9317, as well as excellent ROC-AUC of 0.9859 — effectively equal to the full features baseline model but on a smaller set of features.

6.5 Challenges

- A number of physically implausible blood-pressure and anthropometric records had to be identified and removed before any modelling could begin, requiring careful, clinically-informed cleaning rules rather than a purely statistical outlier filter.
- Because cholesterol and glucose are already meaningfully ordered integer codes rather than free-form categories, a design decision was required on whether to treat them as ordinal (passthrough) or nominal (one-hot) features; passthrough was chosen but this assumption is worth revisiting.

6.6 Future Work

●Conduct SHAP or partial dependence analysis on the finalized Gradient Boosting model to gain deeper insight into the combined influence of pulse pressure and the age-cholesterol interaction.

●Experiment with other ensemble techniques such as XGBoost, LightGBM, or Random Forests along with an increased hyperparameter tuning budget for Gradient Boosting since the current search is rather limited (only 8 RandomizedSearchCV executions).

7. Discussion

7.1 Model Performance

The three classifiers all showed test accuracies and F1-scores above 0.92, with Gradient Boosting proving more effective than the others, Logistic Regression being the most interpretable classifier, while the Neural Network performed quite well but could not dominate. Since there is a rather small difference in the performance of the two different classifiers, one can conclude that after the variables such as Blood Pressure, Cholesterol Level, and Age have been taken into consideration, most of the class separation information in the data is almost linearly separable.

7.2 Impact of Hyperparameter Tuning and Feature Selection

The results of hyperparameter tuning were minor, specific to models, and consisted in increasing Logistic Regression's F1-score slightly (0.9242 baseline → 0.9245 tuned), as well as increasing the cross-validated F1 of the Gradient Boosting algorithm from the baseline training F1 score of 0.9351 to a more conservative (correctly) cross-validated one of 0.9309, and then to a testing F1 score of 0.9317 after choosing proper parameters. Then, feature selection proved to be of actual use in the case of Gradient Boosting by achieving the same results using 6 fewer features than the baseline approach; meanwhile, the performance of Logistic Regression did not change at all.

7.3 Interpretation of Results

The alignment of results from Mutual Information and permutation importance in variables such as pulse pressure, systolic blood pressure, cholesterol, interaction between age and cholesterol, and age itself, along with the EDA observation that increased age and high blood pressure can clearly be observed to be related to disease status, make up a cohesive story: the cardiovascular disease prediction signal in this data set lies in measurements of blood pressure and age/cholesterol risk factors.

7.4 Limitations

●Clarification on the 'CV Score' column of Table 8 and where that data is from: in the case of the baseline models, this refers to the F1 on the training set instead of true cross-validated scores, whereas in the case of the hyperparameter tuned models, this refers to the highest cross-validated scores outputted by GridSearchCV/RandomizedSearchCV – an inconsistency in labels stemming from the comparison table construction is noteworthy here.

●This is not a longitudinal dataset but a one-time cross-section of patient exams, meaning that the model can only detect statistical associations of feature variables and the presence of cardiovascular disease.

7.5 Suggestions for Future Research

●Assess the validity of the final model using an independent cardiovascular data set, as a way of validating whether the chosen features, along with their relative importance, are transferable outside this particular patient sample.

●Consider the use of cost-sensitive cut-offs with the final classifier, considering that in the clinical context, the cost of a false negative (i.e., missing a positive case) is usually much more costly than the false positive.

8. References

Ulianova, S. (2019) Cardiovascular Disease dataset. Kaggle. Available at: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset (Accessed: 2026).

United Nations (2015) Transforming our world: the 2030 Agenda for Sustainable Development, A/RES/70/1. Goal 3, Target 3.4 (Non-communicable diseases).

Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', Journal of Machine Learning Research, 12, pp. 2825–2830.

Pandas, NumPy, Matplotlib and Seaborn — open-source Python libraries used for data processing and visualisation.